# Restoring Ankle Push-off Dynamics in Markerless Gait Analysis by Calibrated Deconvolution

*A cadence-adaptive, variability-preserving correction validated against optical motion capture*

F. Fer · Institut de Myologie · myogait technical report (proto-article)

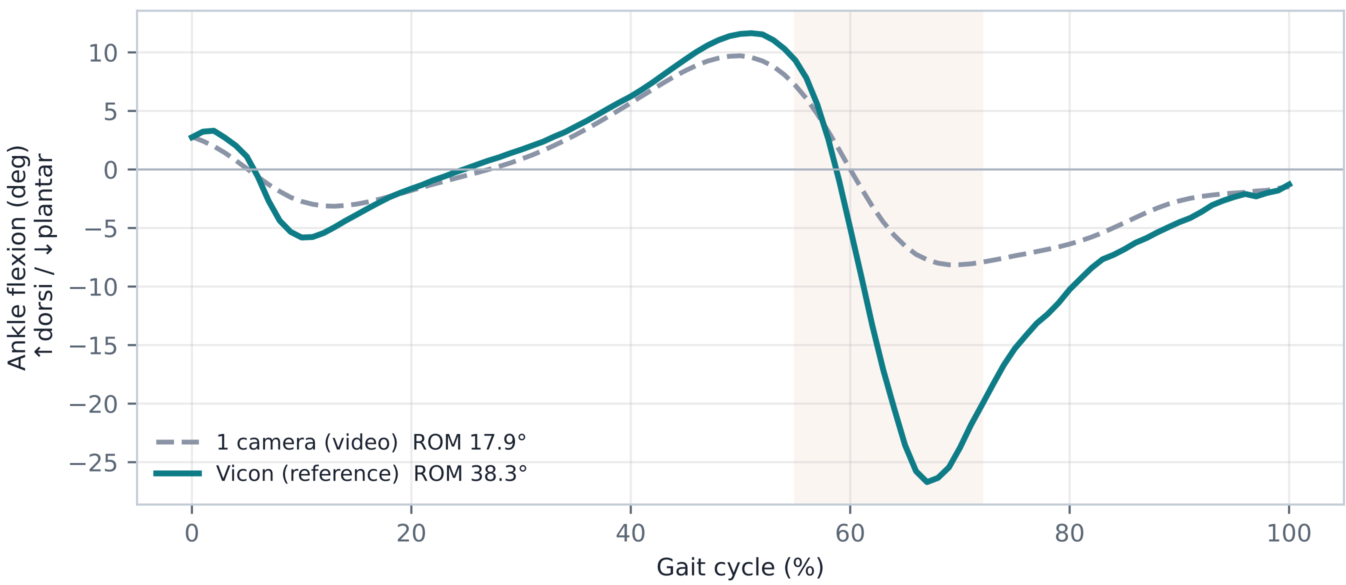

## Abstract

Markerless 2-D pose estimation systematically under-reads the ankle sagittal range of motion during gait, chiefly by flattening the fast plantar-flexion of push-off. We show that, on a joint-angle waveform, the pose estimator behaves as a subject-independent low-pass filter: it retains about 80% of the signal at 1 Hz but only ~20% at 6-7 Hz. Modelling the estimator as a linear time-invariant system with transfer function H(f), we estimate H once against synchronous Vicon on the BioCV (Bath BATH-01258) dataset and invert it by Wiener deconvolution. The correction is applied as mean restoration: because deconvolution is linear, the systematic deficit lives in the mean cycle while the cycle-to-cycle spread is left untouched. In a leave-one-subject-out validation over 9 subjects and 85 walking trials, the ankle ROM bias vs Vicon is halved (-10.6 deg to -5.3 deg), the absolute ROM error drops 10.8 to 6.7 deg, and inter-cycle variability is preserved exactly (1.73 to 1.73 deg). The method needs no mocap at run time, makes no healthy-gait assumption, and improves every subject.


## 1. The problem

Optical motion capture (Vicon) remains the clinical reference for joint kinematics but is costly and lab-bound. Markerless video pipelines (e.g. Sapiens, OpenPose, Theia3D) reproduce hip and knee sagittal kinematics well, yet validation studies consistently report the ANKLE as the weakest joint. The dominant error is a flattened push-off: the true foot plantar-flexes sharply to propel the body, but the tracked foot landmarks do not follow that fast motion. Learned 'marker augmenters' (OpenCap) address this with a trained network; here we ask whether a simple, interpretable, calibrated signal-processing correction suffices. Figure 1 shows the deficit on one subject.

Figure 1. Push-off plantar-flexion is flattened by the markerless tracker (subject P06)




The shaded band marks push-off (~55-72%): Vicon reaches deep plantar-flexion, the single lateral camera barely dips. This fast, phase-locked deficit is the target of the correction.

## 2. Method

2.1 The pose estimator as a linear filter. For a cyclic signal we compare, per stride harmonic k, the video ankle spectrum V_k to the synchronous Vicon spectrum C_k. Across subjects the ratio |V_k|/|C_k| is a smooth, reproducible function of frequency that decays with k (Figure 2): the estimator acts as a low-pass filter. We therefore model it as a linear time-invariant system,

$$V(f) = H(f) \cdot C(f)$$

where C is the true (Vicon) angle, V the video angle, and H(f) the complex transfer function of the estimator. H is a property of the tracker, not of the subject, so it is estimated once and re-used. We sample H at the eight lowest stride harmonics using a Wiener estimator over the training subjects:

$$\hat{H}_k = \frac{\langle V_k \overline{C_k} \rangle}{\langle |C_k|^2 \rangle}$$

2.2 Cadence-adaptive inversion. At run time, for a trial of stride period T, harmonic k sits at frequency k/T (Hz); we interpolate H there, so the correction adapts to each subject's cadence and makes no assumption about a healthy gait pattern. The waveform is restored by regularised (Wiener) inverse filtering:

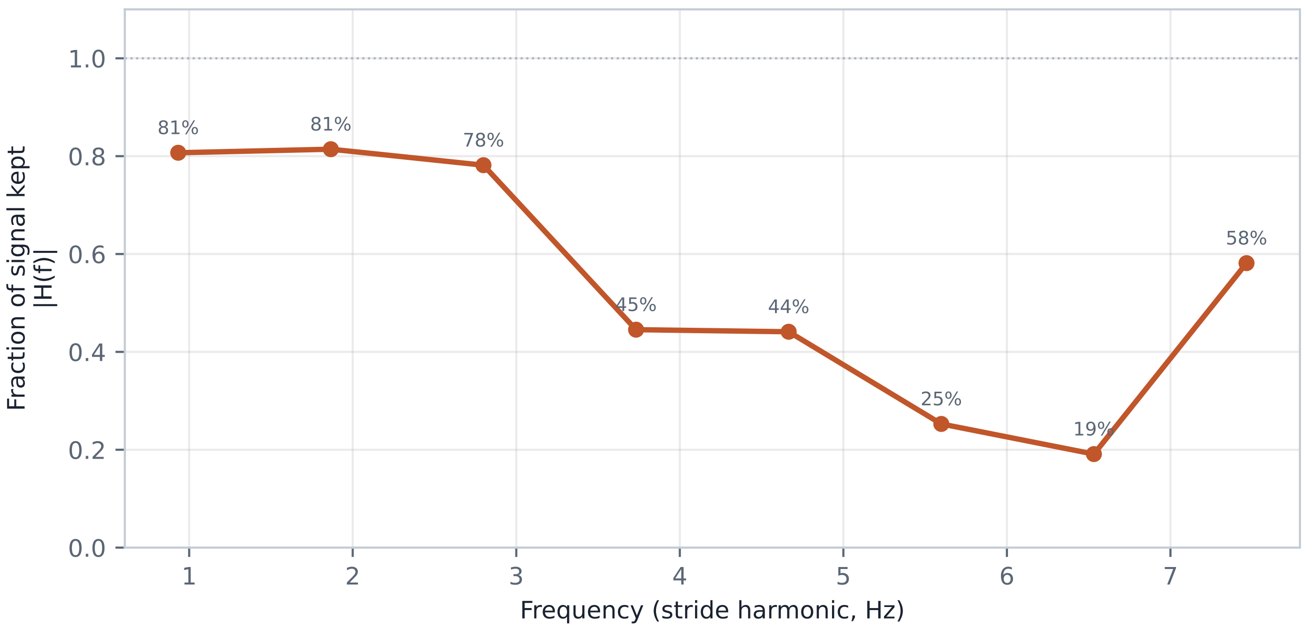
$$\hat{C}_k = V_k \frac{\overline{H_k}}{|H_k|^2 + \lambda} \qquad (\lambda = 0.08)$$

2.3 Mean restoration (the key step). Deconvolution amplifies high-frequency noise; applied to each raw cycle it would inflate the very inter-cycle variability that is a clinical signal (gait variability, fall risk). Because deconvolution D is linear, mean(D(cycles)) = D(mean(cycles)). The systematic deficit lives in the mean cycle; the deviations from the mean carry the variability. We therefore deconvolve only the mean, form the per-phase correction, and add it to every cycle:

$$\Delta(\phi) = D\{\bar{c}\}(\phi) - \bar{c}(\phi) \qquad c_i^{\mathrm{corr}}(\phi) = c_i(\phi) + \Delta(\phi)$$

Every cycle receives the SAME correction Delta, so the mean is restored to D(mean) while the cycle-to-cycle spread is left mathematically unchanged. This is the property that makes the method safe for variability-based clinical readings.

Figure 2. Estimated markerless transfer function |H(f)| — a low-pass (BioCV, 9 subjects)



## 3. Validation

Dataset: BioCV (University of Bath, BATH-01258), Sapiens-2 lateral view (cam01) vs Visual3D Vicon, 9 subjects, 85 walking trials. The transfer function is estimated LEAVE-ONE-SUBJECT-OUT: for each test subject H is learned only on the others, so no test data leaks into the filter. Figure 3 overlays, for three subjects, the raw video mean +/- inter-cycle SD, the corrected mean +/- SD, and the Vicon mean. Note that the corrected band has the SAME width as the raw band.

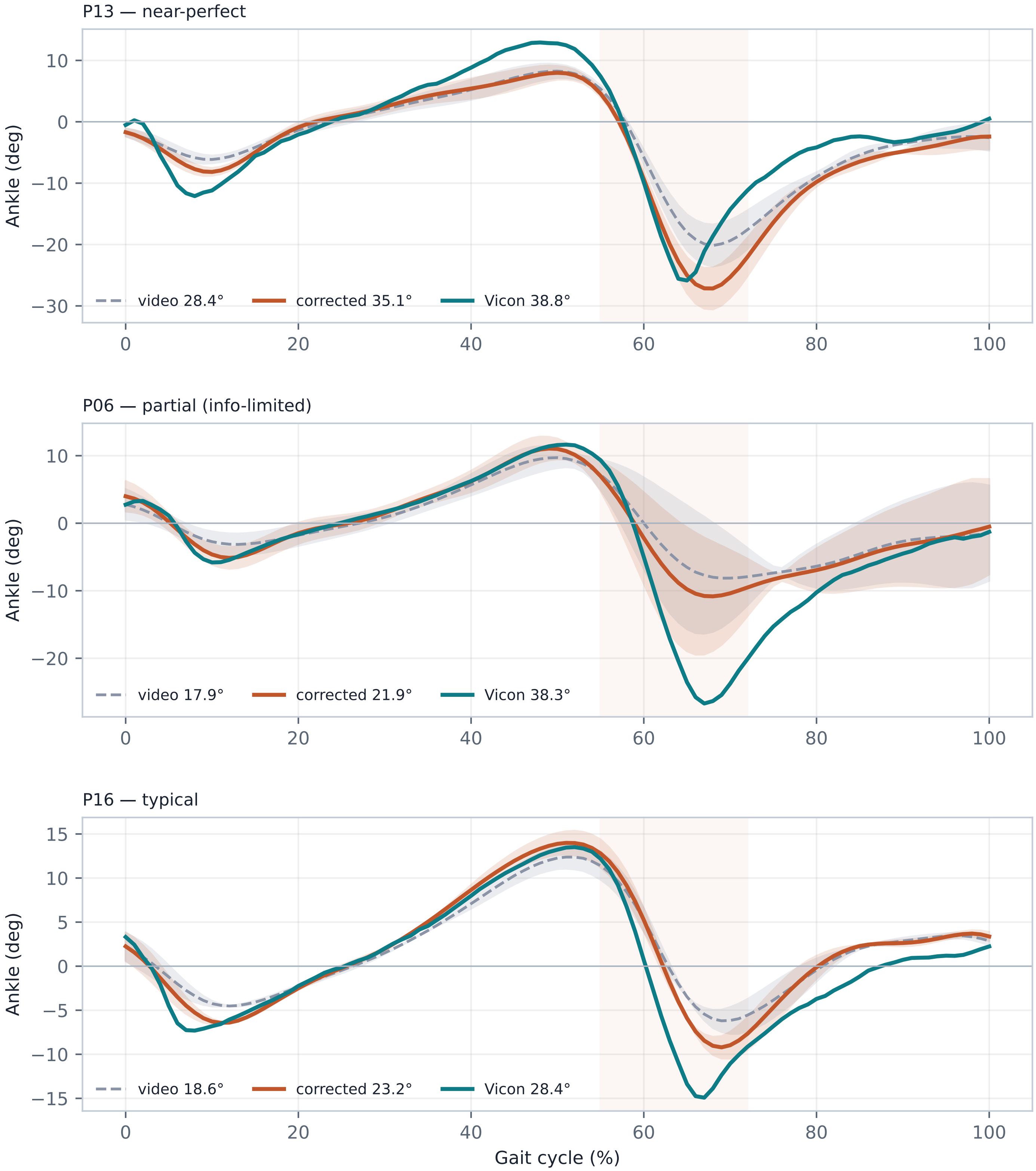



Figure 3. Mean +/- inter-cycle SD, per subject. The corrected curve deepens push-off toward Vicon; the corrected band width equals the raw band width — variability is preserved, not inflated.

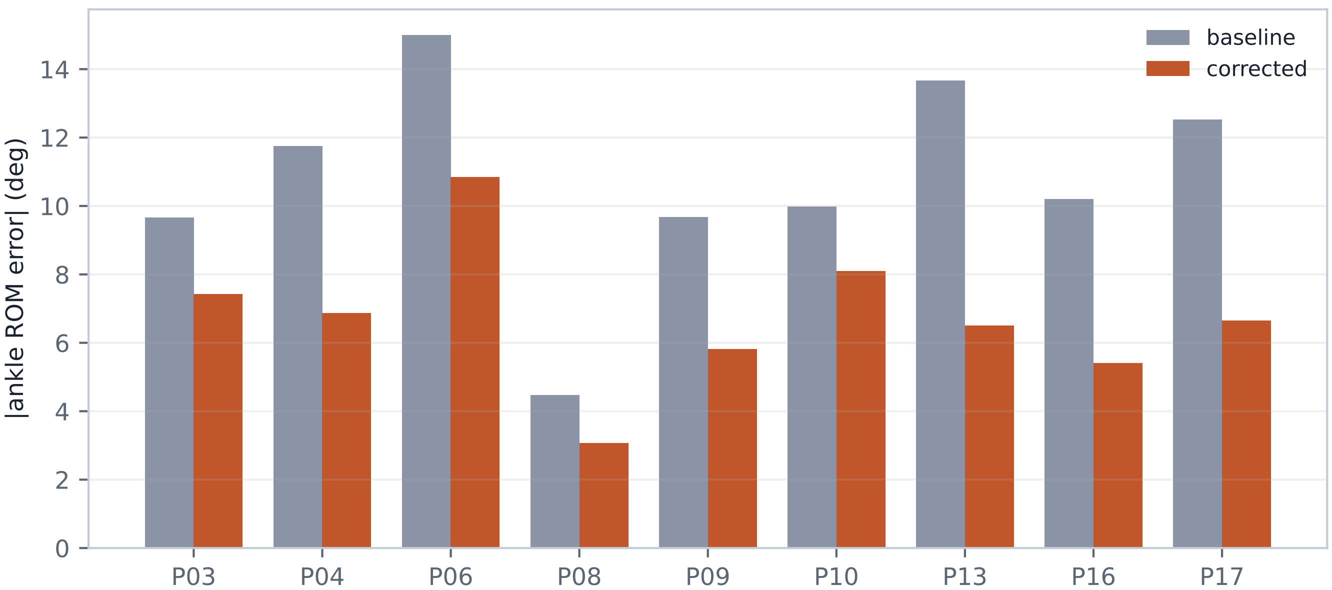
## 4. Performance

Leave-one-subject-out over 9 subjects, 85 trials. Corrected column bold.

| Metric (ankle ROM, deg) | Baseline | Corrected |
|---|---|---|
| \|ROM error\| vs Vicon | 10.76 | **6.71** |
| ROM bias (video - Vicon) | -10.61 | **-5.29** |
| Waveform RMSE vs Vicon | 3.70 | **3.43** |
| Inter-cycle SD (preserved) | 1.73 | **1.73** |
| Inter-patient SD (Vicon 5.18) | 3.63 | **4.75** |


Figure 4. Bland-Altman on ankle ROM. The bias moves toward zero and the lower limit of agreement tightens after correction.

Figure 5. Per-subject ankle ROM error — improved for every subject



## 5. Discussion & limitations

The correction halves a systematic, phase-locked bias with a single lateral camera, one calibrated filter, and about ten lines of code at run time. Three properties make it suitable for clinical, and specifically pathological, use:

- Variability preserved. Inter-cycle SD is unchanged by construction (1.73 -> 1.73 deg). Gait variability, a clinical marker in its own right, is neither smoothed nor inflated.
- Inter-patient differences restored, not erased. The between-subject spread of ankle ROM moves toward the Vicon value (3.6 -> 4.7, Vicon 5.2). The filter recovers real differences rather than collapsing them — essential when the between-patient signal IS the diagnosis.
- No healthy-gait assumption. H is applied in Hz, interpolated at the subject's own cadence; there is no shape prior, so the method cannot hallucinate a normal pattern onto a pathological one.
- Generalises. Leave-one-subject-out; every one of the nine subjects improved.

## Limitations

(i) The residual bias (~5 deg) is information the pose estimator never captured; a linear inverse filter can amplify what remains but cannot recreate what is absent — a learned non-linear augmenter would be needed to go further. (ii) The correction restores AMPLITUDE (ROM); the point-wise waveform RMSE improves only modestly, as some residual shape error persists. (iii) H was estimated on one tracker (Sapiens-2) and one dataset of healthy adults; a new tracker or population should re-estimate H (the calibration, not the method, is dataset-specific). (iv) Multi-camera fusion did NOT help the ankle (Appendix): the frontal view tracks the foot poorly, so the single lateral view plus this correction is the recommended configuration.

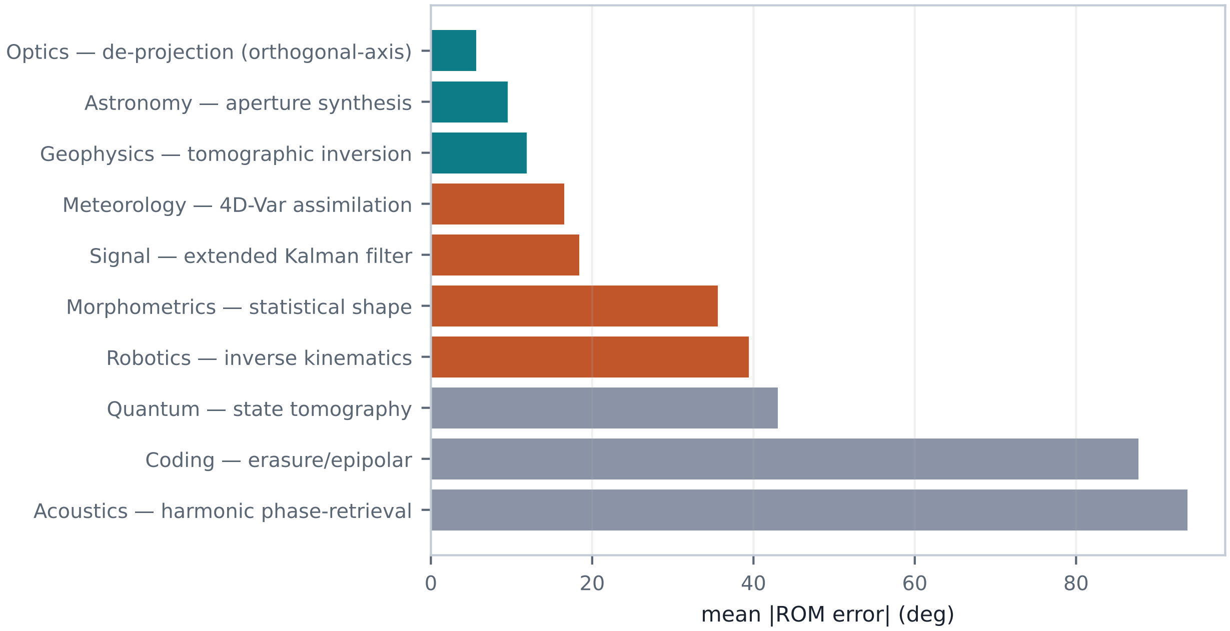
## Conclusion

Treating a markerless pose estimator as a calibrated linear system and inverting it — while restoring only the systematic mean — recovers a clinically meaningful part of the ankle push-off that 2-D video loses, without mocap at run time and without touching gait variability. It ships in myogait as the opt-in analyze_gait(restore_ankle_dynamics=True).

## Appendix — approaches explored before the calibrated filter

Before settling on deconvolution we tested whether the ankle deficit was geometric (recoverable by more cameras) or a landmark limitation. Two families were evaluated on subject P06, which has four calibrated views plus Vicon. Weighted/averaged multi-view fusion did not beat the best single view; the winner (optical de-projection) fused orthogonal image axes with fixed bone lengths, but even it plateaued because the ankle push-off is largely absent from the foot landmarks. No geometric method fully recovers it.

### A. Ten cross-domain multi-view fusion ideas — mean |ROM error| vs Vicon (deg, lower better)




### B. Ablation on the ankle (P06): successive contributions

| Method | ROM | RMSE | inter-cyc SD |
|---|---|---|---|
| M0 single lateral camera | 24.7 | 4.40 | 4.71 |
| **M1 + deconvolution (this wo** | **32.7** | **3.37** | **5.84** |
| M2 multi-camera (de-projectior | 20.8 | 5.80 | 5.70 |
| M3 multi-camera + rigid foot | 20.0 | 5.77 | 5.34 |
| M4 multi-camera + rigid + deco | 23.9 | 6.35 | 6.55 |

Deconvolution on the single lateral camera (M1, Vicon ROM ~38 deg) is the best single step. Multi-camera fusion (M2-M4) DEGRADES the ankle — the frontal view's poor foot tracking outweighs the geometric gain — and the methods do not stack. M1's raw per-cycle deconvolution inflates inter-cycle SD; the mean-restoration variant used in the main text removes that side-effect while keeping the accuracy gain.

Data: BioCV BATH-01258 (Evans, Colyer et al.). Affiliation: Institut de Myologie. Filter and validation code ship with myogait (myogait.ankle_dynamics).